Understanding the earthquake research

A plain language guide to the questions, findings and limits

Open research companion - completed comparisons, 12 November 2026

Complete joint fits, a moderate-thinning control, a stronger ETAS benchmark and block uncertainty are included. Tested attenuation does not improve observed forecasts; these statistical models do not identify physical stress shadows.

Can a model learn when an earthquake makes later activity less likely, as well as when it makes later activity more likely? And does that extra detail actually improve forecasts?

That is the central question of this research. An earthquake catalogue records events: when they happened, where they were located and their reported magnitudes. It does not directly label how much each event encouraged or discouraged later earthquakes. The research asks what can be learned from those records, what requires extra assumptions, and what survives a predictive test.

The main finding

None of the tested attenuation additions improves observed held-out forecasts over its matched excitation model. Large synthetic gains disappear when both models fit the temporal backbone. The moderate control has a small, uncertain mean gain. These results do not show that physical inhibition never happens.

Three different achievements

A model can be mathematically well defined, have parameters that could in principle be distinguished, and still fail to improve predictions from a finite catalogue. The manuscript treats these as separate questions. A good fit to the past is not enough to settle any of them.

The contribution is therefore broader than a league table of forecasts. It develops restricted mathematical results, tests where learning becomes difficult, and records both the successes and failures of particular forecasting procedures.

How to read this guide

Pages 2-4 explain the models and the tests. Pages 5-6 explain the completed simulation and regional findings. Pages 7-8 set out the remaining limits, define common terms and point to the relevant manuscript chapters. No equations are needed.

These are research comparisons using historical catalogues and simulations. The project has not established an operational earthquake forecasting service.

02 The ingredients

What goes into a model

Here, a model is a recipe for assigning a changing rate of earthquake occurrence across time and, in the spatial studies, across a region. It gives a distribution of possible activity rather than a single certain date and location for the next earthquake.

Background

Activity not assigned to earlier recorded events

Triggering

Earlier events raise the modelled rate

Inhibition

A proposed reduction in part of the rate

Different models combine these ingredients in different ways.

Background activity

This is the part of the model's event rate that is not attributed to triggering by recorded earlier events. It may change over time and place. Calling an event a background event is a modelling description; it is not a direct observation of its geological cause.

Triggering and inhibition

Triggering raises the modelled rate after earlier events. Inhibition lowers a specified part of that rate. In the original local-inhibition model, only the background contribution is reduced. One newer version also reduces the triggered contribution. Those are different hypotheses, even when they use the same event history.

The place and the size of an event matter

The spatial models distinguish a region's background pattern from the pattern of later events around an earlier earthquake. A map of long-term activity and a model of an aftershock cloud do different jobs. The research tests choices about both.

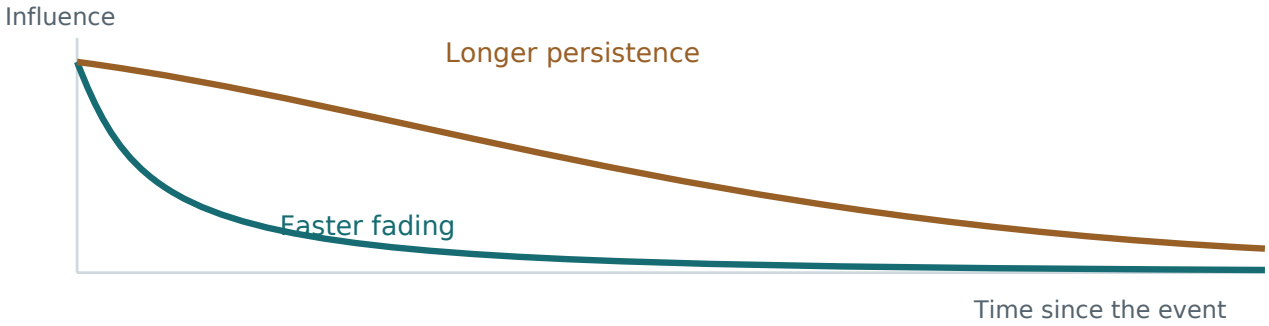
The original local-inhibition model gives all eligible past events the same suppressing weight. LI+, the revised family, gives larger magnitudes more weight. This changes how event size enters the hypothesis. It does not by itself make the model more accurate or turn it into a measured map of stress inside the Earth.

A separately executed constant-background fit links the stability model to the data. It uses the same intensity family as an earlier sensitivity comparison, with a different fitting recipe. Inhibition loses in the early and late periods and is switched off in the middle. The theorem does not establish practical equilibration over the catalogue period.

03 The mathematical questions

Memory and what can be learned

Memory means that the influence of an earlier event fades over time. A short memory concentrates influence soon after an event. A longer memory allows it to persist. It is a feature of the model, not an extra quantity directly recorded in the catalogue.



Schematic only. These are not fitted curves or earthquake observations.

Many simple fading effects can represent a complex memory

Some computations combine a finite collection of simple fading components, each with its own speed. This gives the computer a compact state to update as events arrive. It can approximate a more complicated memory law over a chosen range, but a finite approximation is not the same mathematical object as an exact long-memory law.

Stability has a precise and limited meaning

For the autonomous finite-mode model, the results require constant loading, fewer than one directly triggered event per event on average, and suitable conditions on the response and event properties. Under those assumptions, the theory controls expected activity. That does not mean every simulated sequence is small or that all measures of variability are finite. The manuscript shows why first-moment results - results about averages - can coexist with difficulties involving second moments, which concern variability.

Some settings cannot be distinguished

Imagine turning the strength of a model's inhibitory effect to zero. Changing its proposed range or duration then makes no difference to the forecast. The catalogue cannot identify those unused settings from that forecast. A reported range at zero strength is a convention, not a discovered physical length scale.

Even away from zero, two effects can compete to explain the same pattern. A changing background can make it harder to learn an inhibition effect. Knowing that parameters are distinguishable in an ideal mathematical law does not guarantee accurate estimates from the particular catalogue available.

04 Fit, prediction and calibration

How the comparisons work

Fit first, then test on later observations

The model's adjustable settings are fitted using a training history. A predictive comparison then scores observations outside that fitting period under the declared rules. The relevant comparison uses the same events, time window and region for both models.



Illustrative split. Window lengths and study designs vary.

A score rewards useful probabilities

The research uses log scores: a way of evaluating the probabilities or rates a model assigns to what actually happens. A model cannot win simply by announcing high activity everywhere. Its score also accounts for the activity it predicted across the full observation window. Higher scores are better within the same scoring comparison.

Scores can be separated into contributions for timing, location and magnitude. A model may improve where it places activity while making the timing worse. The overall result and the component results therefore need to be read together.

Calibration asks a different question

A forecast that assigns a one-in-five chance to an event should, across suitable repeated comparisons, see that event about one time in five. That is the idea of calibration. In this research, simulated futures are compared with the actual future to look for discrepancies in counts, event spacing and locations.

Winning does not mean passing every test

One model can score better than another while both give poor descriptions of important features of the data. The benchmark's score advantage is not a calibration certificate.

A log-score difference reported in nats is a difference on a logarithmic scale. It is not a percentage improvement or a number of earthquakes correctly predicted.

05 Experiments with known rules

What the controlled studies show

Simulations create catalogues from specified rules. They let a comparison distinguish a known generating effect from the ability of a particular fitting procedure to recover it.

Knowing the background can change the answer

The historical study compares supplied laws and estimated background parameters on 1,000 no-inhibition histories and 150 alternatives. Detection falls when the background must be estimated. The magnitude-history study likewise shows that matching how magnitude affects triggering reduces apparent evidence for inhibition. A positive fitted strength alone is not a detection.

A fair comparison removes the broad, long-lasting advantage

All 80 original histories now have complete results, giving 400 procedure scores on shared histories. Both models fit their background and temporal-triggering parameters using training data. Matched broad, long-lasting background-only inhibition then loses 10.3701 nats on average, with 10/20 wins. Total-rate inhibition loses 1.2607 nats, with 8/20 wins. These are not 400 independent experiments.

Apparent gains depended on a frozen comparison rate

Refitting only the comparison model's background intercept greatly reduces the original gains. Also fitting its branching ratio reduces the broad/long total-inhibition residual from +9.7367 to +5.6255 nats; all 20 histories choose zero branching. Fitting the full temporal backbone reverses that residual. Wins by a procedure that selects zero inhibition are not detections.

An intermediate regime avoids nearly empty histories

Twenty fresh moderate-thinning histories reject 27.21% of proposed events and retain 94.7 test events on average. The raw mean gain is +0.2941 nats, much smaller than its Monte Carlo standard error of 2.7773; 13/20 histories win. A training-only complexity gate gives +2.4946 nats, with 9/20 wins and positive inhibition in 12/20 histories. Neither row demonstrates reliable recovery.

The moderate control uses a declared score-convergence rule. Eighteen fits do not pass the stricter inherited score-and-gradient comparison, and 13 reach a parameter search boundary. These diagnostics remain visible in the manuscript. Narrow/short immigrant controls thin almost nothing: all 20 original unit-mark paths and 19/20 magnitude-scaled paths reject no events.

06 New Zealand, California and Japan

What the regional studies show

New Zealand: most of the loss is in the first period

Shared inhibition loses 17.62 nats in total against its matched excitation model; about 94% comes from the first test period. Fitted trend and depletion trade off strongly. The inhibited model predicts less integrated activity in that period and loses chiefly at event times. It is not a simple story of monotonically accumulated stress causing rate overprediction.

Paired calendar-block intervals retain dependence within 90-, 180- or 365-day blocks. The tiny second-period contrast remains compatible with zero at each block length. The third-period contrast is sensitive to block length and excludes zero under annual blocks. Small losses cannot all be called indistinguishable from zero.

California: strengthening excitation still selects no attenuation

The original unweighted inverse-power benchmark, EP, beats the background-weighted recipe, OC, in each primary California fold, chiefly through location. A separate five-episode backtest has one reversal. Original and magnitude-weighted background-only inhibition select zero strength in all three primary folds. Raw total-rate inhibition loses less than two nats in each fold.

A stronger ETAS procedure uses magnitude-scaled offspring distances, an immigrant-weighted background and joint fitting, and normalizes offspring on the full plane while retaining regional leakage. Together these changes gain 69.553 nats over the earlier EP procedure. They do not isolate one cause of the improvement. Its matched rupture-scaled attenuation addition selects zero strength in all three folds.

Compare scores with their uncertainty

For the stronger excitation procedure, a paired 90-day interval for the total improvement is [32.239, 123.719] nats. For the smaller original total-inhibition losses, pooled percentile intervals are negative, while paired sign-flip conclusions vary with block length. These calculations hold fitted parameters fixed; they do not include parameter-estimation uncertainty.

Japan exposes a timing problem

The timing advantage over a fixed triggering-time reference in California becomes a loss in Japan. Holding the background trend constant reduces but does not remove that deficit. OC also remains behind EP jointly. Across regions, score improvements and failures concern specific procedures on repeatedly examined historical data, not independent prospective confirmations.

07 The boundaries of the evidence

Interpreting the evidence

A failed model does not rule out a physical process

A model can lose because its form is too restrictive, its background is inadequate, its fitting search misses better settings, the catalogue contains too little information, or the proposed effect adds little predictive value in that setting. These studies narrow some explanations. They do not identify a unique geological cause or prove the absence of physical inhibition.

Looking back is different from issuing a forecast

Using only earlier data when fitting is an important safeguard. It does not turn a retrospectively chosen set of episodes into a genuinely prospective study. Overlapping training histories and related earthquake sequences also limit how many independent pieces of evidence the comparisons supply.

Readers can now rebuild the study inputs

Original provider records and selection rules are now available for the New Zealand reconstruction, California and Japan. Replays rebuild their study catalogues from those records. A fresh run of the frozen New Zealand comparison also reproduced its fitted parameters, event-level scores and numerical controls exactly. This makes those calculations inspectable and repeatable; reused observations remain part of model development. Later CPU runs also recalculated the declared historical comparison scores from the original saved model settings. Their records distinguish rebuilt numerical inputs from exact saved inputs. These checks repeat the scoring, not the original model searches, and do not turn reused observations into new test data.

Better scores do not settle calibration

The completed study compares OC and EP with 999 simulated futures at each of three forecast origins. Combined checks using continuous timing and spatial statistics flag discrepancies for both models in every origin. EP can score better while still giving an inadequate description of some features. These checks hold fitted parameters fixed; they do not include uncertainty about those parameters.

What the supplied records allow

The supplied histories, fitted states and code let readers inspect the completed comparisons. Original California checks covered every simulated path for the declared score fields, with separate complete-vector checks on the selected subset. Later packaging tests are identified separately from those original executions.

The F continuation has been removed because required original fitted states could not be recovered. Its older sources and numerical records remain archived; the supported earlier nuisance study is retained.

08 Finding your way through the manuscript

A small glossary and reading map

Catalogue	The recorded times, locations and magnitudes of selected earthquakes.
Parameter	An adjustable model setting, such as the strength or duration of an effect.
Inhibition	A mechanism that reduces a specified part of the modelled event rate.
Identifiability	Whether different settings could be distinguished from the model's observable law, in principle.
Null control	A simulated case without the effect the procedure is supposed to detect.
Oracle calibration	A simulation-based reference using the known generating law.
Held-out data	Observations outside a particular fitting step; repeated inspection can still make them part of model development.
Zero-strength boundary	The setting where inhibition is switched off and its unused range and memory cannot be identified.
Calibration	Agreement between a forecast's probability statements and repeated outcomes under the declared comparison.
OC, EP, LI and LI+	OC uses a regional background map to weight where triggered activity is placed; EP omits that weighting and uses its own distance-decay rule. LI adds local background suppression to OC; LI+ gives larger events stronger suppressing weights.

Where to read more

The model and its mathematical limits	Chapters 2-5
Study design and controlled detection	Chapters 6-7
Historical temporal and magnitude comparisons	Chapters 8-9
Spatial models, regional transfer and backtests	Chapters 10-11
Calibration and local inhibition	Chapters 12-13
Overall interpretation and remaining questions	Chapter 14

Read results with their comparators and assumptions. Start with docs/data.md and reproduction/README.md; the selected-state reader and docs/cpu_reproduction.md document executed Linux routes. The amended reader at experiments/s04b_refined_cpu_reader/README.md supplies an audited Linux library and finite local limits. Its outputs require separate review; the earlier CPU acceptance covers a different route. Readers can inspect the evidence, report errors and propose revisions.